

Cryptocurrency Exchange and Blockchain Practical Report

For this practical, I explored a cryptocurrency exchange platform and reviewed the account registration process, trading interface, payment methods, transaction records, and security features. I also related these activities to blockchain concepts covered in class.

Step 1 – Account Creation and Identity Verification

During registration, the exchange required an email address, password, phone number, and personal details such as my full name, date of birth, and residential address. Before trading, identity verification required uploading a government-issued identification document and, in many cases, taking a selfie to confirm my identity.

Cryptocurrency exchanges perform identity verification. These laws help prevent fraud, money laundering etc by ensuring that users can be identified.

Step 2 – Exploring the Platform

The dashboard displayed my account balance, recent market prices, and portfolio information. The wallet section showed cryptocurrency balances and provided options to deposit, withdraw, or transfer funds. The trading area allowed users to buy, sell, and monitor price charts.

Three cryptocurrencies listed on the platform included:

- Bitcoin (BTC) is the first and most widely recognised cryptocurrency. It is mainly used as a digital store of value and for peer-to-peer payments. Its limited supply of 21 million coins makes it different from many other cryptocurrencies.
- Ethereum (ETH) is a blockchain platform that supports smart contracts and decentralised applications. Unlike Bitcoin, Ethereum allows developers to build blockchain applications and create digital assets such as NFTs.
- USD Coin (USDC) is a stablecoin whose value is designed to remain close to one US dollar. Unlike Bitcoin and Ethereum, its primary purpose is price stability rather than investment growth.

Step 3 – Adding a Payment Method

The exchange allowed users to add either a bank account or a debit card. A bank account required banking details such as the account number and account holder information, while a debit card required the card number, expiry date, CVV, and billing address.

Step 4 – Simulating a Purchase

I simulated purchasing Ethereum (ETH) with \$100. A market order purchases cryptocurrency immediately at the current market price, while a limit order only executes if the market reaches a price chosen by the buyer.

I selected a market order because it provides immediate execution. The amount of Ethereum received depends on the current market price and any transaction fees charged by the exchange. The order preview displayed the estimated amount before confirming the purchase.

Step 5 – Transaction History

The transaction history records the date and time of each transaction, the cryptocurrency involved, transaction type (buy, sell, deposit, or withdrawal), amount, fees, and transaction status.

Step 6 – Security Features

One important security feature is two-factor authentication (2FA), which requires a second verification code in addition to the password. This greatly reduces the risk of unauthorised account access.

A **wallet address** is a unique blockchain identifier used to send and receive digital assets. It represents ownership of cryptocurrency or tokens on the blockchain.

A **smart contract** is a program stored on the blockchain that automatically executes predefined rules. Unlike a traditional server application, its code and data are decentralised and cannot be modified after deployment without following blockchain rules.

An **NFT (Non-Fungible Token)** represents ownership of a unique digital asset. The ALU logo was registered using the ERC-721 standard because every registered logo is unique and cannot be exchanged on a one-to-one basis like ordinary tokens.

A **SHA-256 hash** is a cryptographic fingerprint of a file. Changing even a single pixel in the logo produces a completely different hash, making it easy to detect any modification. Once the hash and asset information are stored on the blockchain, they cannot be altered because each block is cryptographically linked to previous blocks and validated by the blockchain network.

Tokenisation

Tokenisation is the process of representing ownership of an asset using digital tokens. In this project, the ALU logo was first registered as a unique NFT and then ownership shares were represented using ALUT ERC-20 tokens. These tokens could be distributed to faculty members, administrators, researchers, alumni, or other stakeholders to represent fractional ownership or participation in governance.

The ERC-721 standard is used for unique assets such as the ALU logo because every NFT has its own identity. The ERC-20 standard is designed for interchangeable tokens, making it suitable for representing ownership shares that can be divided and transferred between many holders.

If a faculty member owns 100,000 ALUT tokens out of a total supply of 1,000,000, they effectively hold 10% ownership of the tokenised asset. This represents one-tenth of the total token supply and could be used to determine voting rights, rewards, or other benefits if such mechanisms were implemented.